Mahavir Chaudhari

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Experience

Data Engineer II, Providence Global Center - Hyderabad Jul 2025 - Present

- **Designed and implemented** a **financial reconciliation data layer** in **Snowflake** using dimensional modeling, SCD Type 2, and ELT pipelines, consolidating **1B+ records daily** and reducing manual reconciliation effort by **80%.**
- **Built and owned** an AIDLC workflow to automate source discovery, canonical data modeling for a unified data model supporting healthcare applications, and while generating data contracts, compressing a **5-month timeline to 4 weeks** (80% reductions) saving time and cost.
- **Spearheaded** the migration of DSS platform from **Oracle Exadata** to Snowflake's cloud warehouse saving **$1.25M** in cost and leveraged snowflake's warehouse capabilities to reduce runtime down from **15 hours to 12 hours** also leveraging AI to compress timelines by **50% (8 months to 4 months)** .
- **Optimized** Cost Insight's ETL pipelines through parallel DML, higher DOP, partitioning, and configuration-driven Oracle procedures using LLM engineering and reducing runtime by **25% (20h → 15h)** saving $16k in warehouse cost annually.
- **Architected** the **AI Data Life cycle (AIDLC)**, an enterprise agentic AI platform standardizing context engineering, MCP integrations, reusable skills, and agent orchestration with **HIPAA-compliant guardrails** for production AI development and testing, resolving KTLO items instantaneously.

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- **Re-architected** the Mandatory Days Off (MDO) scheduling engine by redesigning data models, query paths, and batch workflows across enterprise warehouses, reducing execution time from **2 hours to 10 minutes (92%)** while supporting scheduling for all hospitals in the region and reducing workload for **6,000+ clinicians**.
- **Modernized** enterprise data infrastructure by migrating **400+** PySpark ETL workloads and 11 production ingestion pipelines from **Databricks to Snowflake**, leveraging AI-assisted automation while achieving **zero production downtime**.
- **Engineered** a metadata-driven AI validation framework using Python, Streamlit, and dynamic SQL, automating reusable data reconciliation across ETL pipelines, reducing **development time by 86% (7h → 1h)**.

Software Development Intern, Fortranix Technologies - Bangalore Jan 2023 - Jun 2023

- **Developed** automated **Rust** unit tests for an enterprise security platform, increasing code coverage from **30% to 80%** and improving production reliability.

Software Developer (Part-time), Streamn - Remote May 2021 - Aug 2022

- **Built** GPU-accelerated computer vision models using **Python, TensorFlow/PyTorch, Jupyter, and Google Colab** to automate logo detection across video streams, improving the scalability of media content analysis.

Projects

Enterprise AI Context Engineering

- **Architected** an **enterprise context engineering platform** with **30+ reusable Data Engineering skills**, MCP integrations, dynamic context composition, semantic retrieval, and tool orchestration incorporating **semantic models, ontologies, knowledge graphs**, and HIPAA-compliant PII/PHI guardrails to standardize enterprise AI agents.

Smart Research Scout

- **Built** a **full-stack AI research platform** using FastAPI, React and vector search to aggregate technical knowledge from research papers, developer communities, and social platforms into a unified **semantic search and analytics interface**.

Technical Skills

- **Data Engineering:** Python, SQL, Snowflake, Azure Data Factory (ADF), Distributed systems, PySpark, dbt, ETL/ELT, Data Modeling, Dimensional Modeling, SCD, Data Warehousing, Batch Processing, Control-M
- **AI Systems:** Agentic AI, Context Engineering, Semantic Layers, Ontologies, Knowledge Graphs, Retrieval-Augmented Generation (RAG), MCP, LangGraph, AI Agents, Tool Calling, Semantic Search, Vector Search, APIs, Orchestration
- **Cloud:** Azure, AWS, GCP, Docker, Git, GitHub Actions
- **Languages & Frameworks:** FastAPI, Flask, React, PostgreSQL, Redis, MySQL, SQLite, Java, Go, Rust, Bash, JavaScript, HTML/CSS

Education

BITS Pilani, BE in Computer Science 2019-2023